REVIEW-1

Title:- Unified Explainable Artificial Intelligence for Business Analytics : Comparing ML Models and XAI Techniques

Problem Statement:

Machine learning models can provide accurate predictions but are often difficult to interpret. Different models and XAI techniques may produce different levels of accuracy and explainability. This project compares **Logistic Regression, Decision Tree, Random Forest, XGBoost and CatBoost** using benchmark educational analytics datasets. **SHAP and LIME** are used to understand and compare model predictions.

Objectives:

- **To develop** machine learning models for analyzing and predicting outcomes from benchmark business datasets.
- **To integrate** SHAP and LIME techniques to provide clear and meaningful explanations of model predictions.
- **To enhance** model transparency and interpretability for better understanding of business-related predictions.
- **To evaluate** and compare model performance and explainability to identify the most effective approach.

Literature Survey:

No.	Paper	Year	Publication	Key contribution
1	**Model-Agnostic Explainable Artificial Intelligence Methods in Finance: A Systematic Review, Recent Developments, Limitations, Challenges and Future Directions**	2025	Springer – *Artificial Intelligence Review*	Reviews SHAP, LIME, PDP, counterfactuals and other model-agnostic XAI techniques in finance.
2	**A Comprehensive Review on Financial Explainable AI**	2025	Springer – *Artificial Intelligence Review*	Reviews XAI techniques, applications and challenges in financial decision-making.
3	**Explainable Artificial Intelligence: A Survey of Needs, Techniques, Applications, and Future Direction**	2024	Survey/Review	Provides a broad taxonomy of XAI methods, applications, requirements and future research directions.
4	**Explainable Artificial Intelligence (XAI) in Finance: A Systematic Literature Review**	2024	Springer – *Artificial Intelligence Review*	Systematically reviews XAI applications in finance and highlights challenges in evaluating explanations.
5	**A Comprehensive Taxonomy for Explainable Artificial Intelligence: A Systematic Survey of Surveys on Methods and Concepts**	2023/2024	Springer – *Data Mining and Knowledge Discovery*	Provides a taxonomy for classifying XAI methods and concepts, useful for comparing different approaches.
6	**Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence**	2024	Springer – *Cognitive Computation*	Reviews techniques for interpreting black-box ML models and discusses their strengths and limitations.

No.	Paper	Year	Publication	Key contribution
7	**Transformative Impact of Explainable Artificial Intelligence: Bridging Complexity and Trust**	2025	Springer – *Discover Artificial Intelligence*	Reviews model-specific, model-agnostic and hybrid XAI methods and discusses accuracy–interpretability trade-offs.
8	**A Review of Evaluation Approaches for Explainable AI with Applications in Cardiology**	2024	Springer – *Artificial Intelligence Review*	Examines how XAI explanations can be evaluated and highlights the lack of systematic evaluation.
9	**Privacy-Preserving Explainable AI: A Survey**	2024/2025	Springer – *Science China Information Sciences*	Reviews XAI methods while considering privacy issues and challenges in explainable systems.
10	**Developing User-Centered System Design Guidelines for Explainable AI: A Systematic Literature Review**	2025	Springer – *Artificial Intelligence Review*	Reviews user-centered XAI design and identifies factors affecting explanation usefulness and user trust.

Research Gap identification:

- No universally best XAI method: Different XAI techniques perform differently depending on the ML model and business problem.
- Limited direct comparison: Many studies focus on individual XAI techniques or specific domains rather than comparing several methods under the same business analytics setting.
- Incomplete evaluation: Existing research often emphasizes prediction accuracy or visual explanations, while interpretability, consistency, computational cost and explanation quality are not always evaluated together.
- Business-specific gap: A significant portion of recent XAI research is concentrated on healthcare and finance, leaving scope for a broader business analytics comparison framework.
- User perspective: Producing an explanation does not necessarily mean that business users can understand or effectively use it. Recent research therefore emphasizes user-centered evaluation.

Proposed Research Gap:

Existing studies have limited unified evaluation of ML models and XAI techniques across multiple business datasets, particularly considering **explanation fidelity, sparsity, interpretability, and computational efficiency** along with predictive performance.

Innovation:

• **To develop** a common framework to compare multiple ML models with XAI techniques across benchmark business datasets.

• **To integrate** fidelity and sparsity measures to assess the reliability, simplicity, and quality of model explanations.

• **To enhance** business decision-making by analyzing the accuracy, fidelity, sparsity, interpretability computational effectiveness.

• **To evaluate** different ML–XAI combinations and recommend the most suitable approach based on prediction performance, fidelity, and sparsity.

Feasibility:

Technical Feasibility-

The project can be implemented using **Python, Scikit-learn, XGBoost, SHAP, LIME, Pandas and Matplotlib/Seaborn**.

Data Feasibility-

Public business datasets such as customer churn, credit risk, sales or marketing datasets can be used.

Economic Feasibility-

The project requires only open-source software and publicly available datasets, making the development cost minimal.

Time Feasibility-

The project can be completed through sequential stages: literature review → data preprocessing → ML models → XAI implementation → comparison → evaluation.

Project Plan

Phase	Work
Phase 1	Literature survey and identification of research gap
Phase 2	Dataset collection and preprocessing
Phase 3	Implementation of ML models
Phase 4	Implementation of SHAP, LIME, PDP and other XAI methods
Phase 5	Comparative evaluation using defined metrics
Phase 6	Analysis, visualization and recommendation framework
Phase 7	Documentation, report and final presentation

BATCH-2

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